# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 10 — MATHEMATICS TWO-CHAPTER TEST SET A

Chapters 8 & 9: **Introduction to Trigonometry, Some Applications of Trigonometry** | RBSE Board Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures for questions on heights and distances. Take $\sqrt{3} = 1.732$ where needed.

### SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** The value of $\sin 45^\circ \cos 45^\circ$ is: 1

**(a)** 1 **(b)** $\frac{1}{2}$ **(c)** $\sqrt{2}$ **(d)** $\frac{1}{\sqrt{2}}$

**2.** If $\cot A = 1$ ($A$ acute), then $A$ is: 1

**(a)** $45^\circ$ **(b)** $30^\circ$ **(c)** $90^\circ$ **(d)** $60^\circ$

**3.** The value of $\cos^2 30^\circ - \sin^2 30^\circ$ is: 1

**(a)** $\frac{\sqrt{3}}{2}$ **(b)** 0 **(c)** $\frac{1}{2}$ **(d)** 1

**4.** A student states: "$\tan 90^\circ = 1$." The correct response is: 1

**(a)** Correct **(b)** $\tan 90^\circ = 0$

**(c)** Incorrect — $\tan 90^\circ$ is not defined **(d)** Cannot be determined

**5.** If $\sin A = \frac{1}{2}$, Student A says $\cos A = \frac{\sqrt{3}}{2}$; Student B says $\cos A = \frac{1}{2}$. Then: 1

**(a)** Student A is correct **(b)** Student B is correct

**(c)** Both are correct **(d)** Both are wrong

**6.** A ladder makes $30^\circ$ with the ground and reaches 3 m up a wall. Its length is: 1

**(a)** $2\sqrt{3}$ m **(b)** $3\sqrt{3}$ m **(c)** 3 m **(d)** 6 m

**7.** When the shadow of a pole equals its height, the sun's elevation is: 1

**(a)** $90^\circ$ **(b)** $60^\circ$ **(c)** $30^\circ$ **(d)** $45^\circ$

**8.** From the top of a 100 m tower, the angle of depression of a car is $30^\circ$. Its distance from the foot is: 1

**(a)** $\frac{100}{\sqrt{3}}$ m **(b)** 100 m **(c)** $100\sqrt{3}$ m **(d)** 200 m

**9.** A student states: "An angle of elevation can be more than $90^\circ$." The correct response is: 1

**(a)** Correct

**(b)** Only at $180^\circ$

**(c)** Incorrect — it lies between $0^\circ$ and $90^\circ$

**(d)** Cannot be determined

**10.** A kite is 50 m high and its string makes $30^\circ$ with the ground. Student A says the string is 100 m; Student B says $50\sqrt{3}$ m. Then: 1

**(a)** Student B is correct **(b)** Student A is correct

**(c)** Both are correct **(d)** Both are wrong

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** If $\sin\theta = \frac{8}{17}$, find $\cos\theta$ and $\tan\theta$. 2

**12.** Evaluate: $4\cot^2 45^\circ - \sec^2 60^\circ + \sin^2 60^\circ + \cos^2 90^\circ$. 2

**13.** A tower is $50\sqrt{3}$ m high. The angle of elevation of its top from a point on the ground is $60^\circ$. Find the distance of the point from the tower. 2

**14.** The sun's elevation is $60^\circ$ and a tower's shadow is 10 m. Find its height. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** Prove that $(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) = 2$. 3

**16.** From a point 20 m from the foot of a building, the angle of elevation of the top of the building is $45^\circ$ and that of the top of a flagpole on it is $60^\circ$. Find the length of the flagpole. 3

**17.** If $\sin A + \sin^2 A = 1$, prove that $\cos^2 A + \cos^4 A = 1$. 3

**OR**

Prove that $(\sin\theta + \cos\theta)^2 + (\sin\theta - \cos\theta)^2 = 2$. 3

**18.** From the top of a 30 m tower, the angles of depression of two points on the ground on the same side are $30^\circ$ and $60^\circ$. Find the distance between the points. 3

**OR**

A kite is flying at a height of 75 m. Its string makes $60^\circ$ with the ground. Find the length of the string. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** Prove that $\frac{\sin A - 2\sin^3 A}{2\cos^3 A - \cos A} = \tan A$. Verify it for $A = 30^\circ$. 5

**OR**

Prove that $\frac{\tan A + \sec A - 1}{\tan A - \sec A + 1} = \frac{1 + \sin A}{\cos A}$. 5

**20.** **Case study:** From the top of a 15 m building, the angle of elevation of the top of a tower is $30^\circ$ and the angle of depression of its foot is $45^\circ$. 5

(i) Draw the figure.

(ii) Find the distance between the building and the tower.

(iii) Find the height of the tower.

**OR**

The angle of elevation of a cloud from a point 60 m above a lake is $30^\circ$ and the angle of depression of its reflection in the lake is $60^\circ$. Find the height of the cloud above the lake. 5

**END OF QUESTION PAPER** — **Total: 40 Marks**